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$$\begin{aligned}\int \sin^5 x dx &= \int \sin^4 x d(-\cos x) \\ t &= \cos x \\ &= - \int (1 - t^2)^2 dt \\ &= -t - \frac{2t^3}{3} + \frac{t^5}{5} + c \\ &= -\cos x + \frac{2\cos^3 x}{3} - \frac{\cos^5 x}{5} + c\end{aligned}$$

References